Experiment 15

Discrete-Time Fourier Transform and Frequency-Domain Analysis

OBJECTIVES

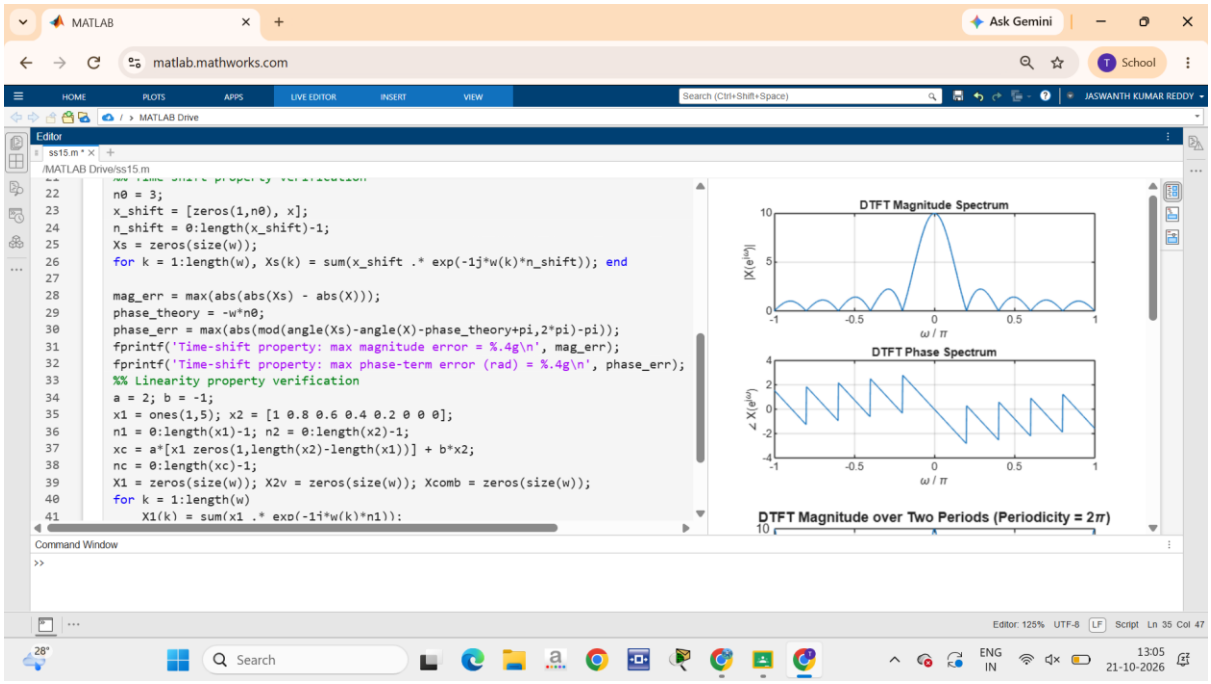
• Compute the DTFT of finite-length discrete-time sequences.

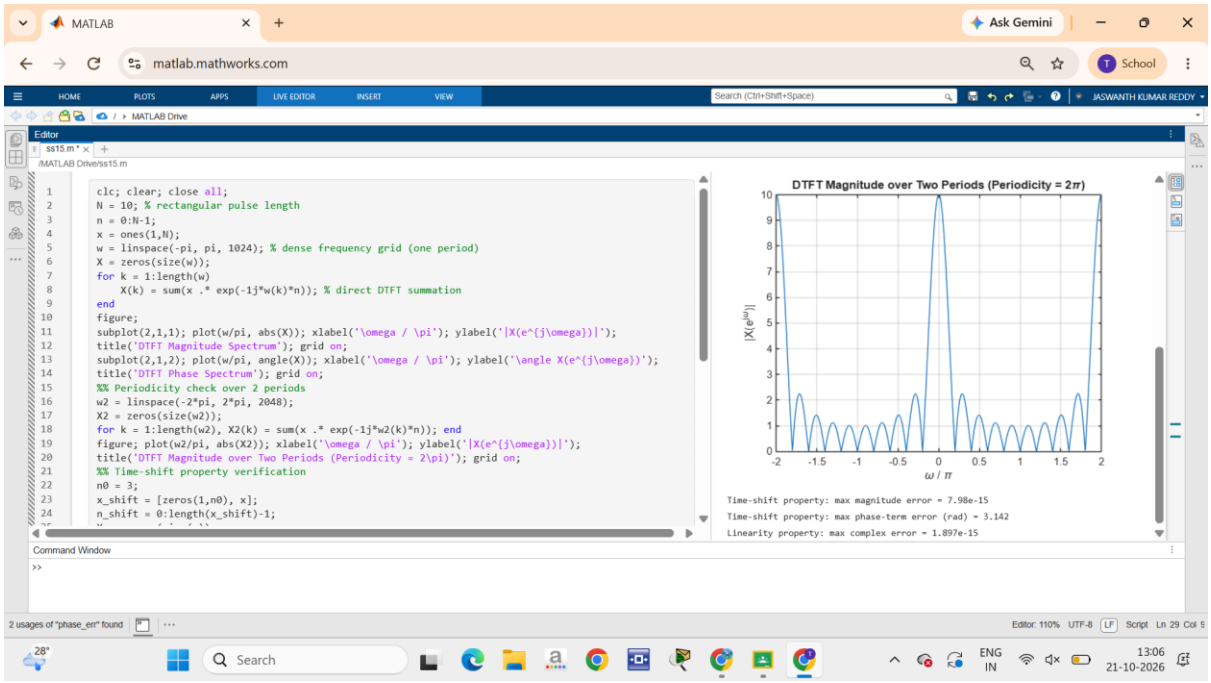
• Plot and interpret DTFT magnitude and phase spectra.

• Verify DTFT periodicity with period 2π and key properties including linearity and time shift.

• Relate sequence length to main-lobe width and spectral spreading.

Output Graph:





Result: The discrete-time Fourier transform (DTFT) was used to analyze discrete-time signals and obtain their frequency-domain representations.